INSTRUCTIONS: Solve each quadratic equation using the most appropriate method (factoring, completing the square, or the quadratic formula), and simplify completely. Write "No Real Solution" if the equation has none.

EXAMPLE:

a) $x^2 - 5x + 6 = 0$ (Factoring)
Factor: $(x - 2)(x - 3) = 0 \rightarrow x = 2$ or $x = 3$.

b) $x^2 + 6x = 7$ (Completing the Square)
Add $(6/2)^2 = 9$ to both sides: $x^2 + 6x + 9 = 16 \rightarrow (x + 3)^2 = 16 \rightarrow x + 3 = \pm 4 \rightarrow x = 1$ or $x = -7$.

c) $x^2 + 2x + 5 = 0$ (Quadratic Formula)
Discriminant: $b^2 - 4ac = 4 - 20 = -16$. Since this is negative, there is no real square root — No Real Solution.

I. SOLVING BY FACTORING

1. x^2 - x = 6

2. 14x - 11x^2 = 3

3. 3x^2 - 18x = 0

4. a^2 + 5a + 6 = 0

5. x^2 - 6x + 9 = 0

6. x^2 + 7 = 0

7. 2x^2 + 3x - 2 = 0

8. 6x^2 - 5x - 6 = 0

9. x^4 - 9x^2 + 8 = 0

10. x - 3√x - 4 = 0

11. 1/(2x - 1) - 1/(2x + 1) = 1/4

12. √(y - 5) + √y = 5

II. SOLVING BY COMPLETING THE SQUARE

13. a^2 - 12a + 11 = 0

14. 2x^2 + 5x = 12

15. 10r^2 = -19r - 6

16. 3a^2 + 7a = 5

17. x^2 + 8x + 5 = 0

18. 2x^2 - 3x - 5 = 0

19. x^2 - 4x = 12

20. 4x^2 + 4x - 3 = 0

III. SOLVING USING THE QUADRATIC FORMULA

21. $3y^2 - 5y = 2$

22. $2a^2 + a - 15 = 0$

23. $6x^2 + 7x - 20 = 0$

24. $4y^2 + 16y + 15 = 0$

25. $x^2 + 4x + 5 = 0$

26. $4x^2 - 12x + 9 = 0$

27. $x^2 - 2x - 1 = 0$

28. $2x^2 + 3x + 5 = 0$

29. $9x^2 - 6x + 1 = 0$

30. $5x^2 - 2x - 4 = 0$

IV. RECOGNIZING THE RIGHT APPROACH

31. $x^2 = 5x - 6$

32. $(x + 3)(x - 2) = 6$

33. $x(x - 4) = -4$

34. $(2x - 1)^2 = 9$

35. $3x^2 - 12 = 0$

36. $x^2 + 5x = 0$

37. $2x^2 + 3 = 2x^2 + 3x$

38. $(x + 1)^2 = x^2 + 2x + 1$

39. $x^2 - 4 = (x - 2)(x + 3)$

40. $3x^2 - 4x - 2 = 0$

Mastery Checklist — Situations You Should Recognize on Sight

Section I — Factoring

- Standard trinomial, $a = 1$
- Trinomial with $a \neq 1$, or one that needs rearranging first
- Incomplete quadratic — common factor only ($ax^2 + bx = 0$)
- Pure quadratic ($ax^2 + c = 0$)
- Perfect square trinomial → repeated (double) root
- Sum of squares (x^2 + positive number = 0) → No Real Solution
- Reducible quartic — substitute $u = x^2$
- Reducible radical equation — substitute $u = \sqrt{x}$ or isolate-and-square
- Reducible rational equation — clear denominators first

Section II — Completing the Square

- Leading coefficient already 1
- Leading coefficient ≠ 1 — divide every term first
- Roots come out rational
- Roots come out irrational — leave the answer in simplified radical form

Section III — Quadratic Formula

- Discriminant is a perfect square → two rational roots
- Discriminant is positive but not a perfect square → two irrational roots
- Discriminant equals 0 → one repeated (double) root
- Discriminant is negative → No Real Solution

Section IV — Recognizing the Right Approach

- Equation not yet in standard form — rearrange first
- Must expand before moving terms — don't set the original factors equal to a nonzero number
- Expanding reveals a perfect square (repeated root)
- (expression)^2 = constant — use the square root property directly, no need to expand
- x^2 terms cancel out entirely — the equation is actually linear
- x^2 terms cancel and leave an identity — Infinitely Many Solutions
- No obvious factors — default to the quadratic formula

Answer Key

1. *// rewrite as $x^2 - x - 6 = 0$, then factor*
x = 3, x = -2

2. *// rearrange into standard form first, then factor*
x = 3/11, x = 1

3. *// factor out the common monomial*
x = 0, x = 6

4. *// find two numbers that multiply to 6 and add to 5*
a = -2, a = -3

5. *// recognize the perfect square trinomial*
x = 3 (double root)

6. *// isolate x^2; a negative square has no real root*
No Real Solution

7. *// factor with a ≠ 1: find factors of 2 and -2 that combine to give 3x*
x = 1/2, x = -2

8. *// factor with a ≠ 1: find factors of 6 and -6 that combine to give -5x*
x = -2/3, x = 3/2

9. *// substitute $u = x^2$, factor, then solve for x afterward*
x = ±1, x = ±2√2

10. *// substitute $u = \sqrt{x}$; reject any negative value of u*
x = 16

11. *// combine into a single fraction, cross-multiply, then solve*
$x = 3/2$, $x = -3/2$

12. *// isolate one radical, square, then isolate and square again*
$y = 9$

13. *// half of -12 is -6; add 36 to both sides*
$a = 11$, $a = 1$

14. *// divide by the leading coefficient before completing the square*
$x = 3/2$, $x = -4$

15. *// rearrange to $ar^2 + br + c = 0$, divide by the leading coefficient, then complete the square*
$r = -2/5$, $r = -3/2$

16. *// divide by the leading coefficient — the roots don't come out rational here*
$a = (-7 \pm \sqrt{109})/6$

17. *// half of 8 is 4; add 16 to both sides*
$x = -4 \pm \sqrt{11}$

18. *// divide by the leading coefficient, then complete the square*
$x = 5/2$, $x = -1$

19. *// half of -4 is -2; add 4 to both sides*
$x = 6$, $x = -2$

20. *// divide by the leading coefficient, then complete the square*
$x = 1/2$, $x = -3/2$

21. *// rewrite in standard form, then apply the quadratic formula*
$y = 2$, $y = -1/3$

22. *// identify a, b, c, then apply the quadratic formula*
$a = 5/2$, $a = -3$

23. *// identify a, b, c, then apply the quadratic formula*
$x = 4/3$, $x = -5/2$

24. *// identify a, b, c, then apply the quadratic formula*
$y = -3/2$, $y = -5/2$

25. *// compute the discriminant first: $b^2 - 4ac$*
No Real Solution (discriminant = -4)

26. *// compute the discriminant first — it comes out to 0*
$x = 3/2$ (double root)

27. *// the discriminant is positive but not a perfect square — simplify the radical*
$x = 1 \pm \sqrt{2}$

28. *// compute the discriminant first: $b^2 - 4ac$*
No Real Solution (discriminant = -31)

29. *// compute the discriminant first — it comes out to 0*
$x = 1/3$ (double root)

30. *// the discriminant is positive but not a perfect square — simplify the radical*
$x = (1 \pm \sqrt{21})/5$

31. *// move every term to one side first, then factor*
$x = 2, x = 3$

32. *// expand and move the 6 over — do not set the original factors equal to 6*
$x = -4, x = 3$

33. *// expand first; this simplifies to a perfect square trinomial*
$x = 2$ (double root)

34. *// take the square root of both sides directly — no need to expand*
$x = 2, x = -1$

35. *// isolate x^2 and use the square root property*
$x = \pm 2$

36. *// factor out the common x*
$x = 0, x = -5$

37. *// the x^2 terms cancel — this is actually a linear equation*
$x = 1$ (Actually Linear)

38. *// expand the left side and compare — both sides turn out identical*
Infinitely Many Solutions (Identity)

39. *// expand the right side — the x^2 terms cancel, leaving a linear equation*
$x = 2$ (Actually Linear)

40. *// this doesn't factor evenly — use the quadratic formula*
$x = (2 \pm \sqrt{10})/3$